

COURSE NOTES

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# TOPICS IN ALGEBRA: GROUP THEORY

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# Chapter 1

## Lecture 1

Lie algebras were originally introduced by Sophus Lie in the 1870s to study the concept of continuous transformation groups. Lie's motivation was to develop a theory for differential equations analogous to Galois theory for algebraic equations. He discovered that the infinitesimal transformations of a continuous group form a linear space equipped with a bracket operation, now known as a Lie algebra. This linearization allows complex global geometric problems to be translated into algebraic problems, making them more tractable.

Let  $F$  be a field (mostly  $\text{char } F = 0$ , even  $F = \mathbb{C}$ ).

**Definition 1.0.1.** *A linear algebra  $A$  is a vector space over  $F$ , with a binary bilinear operation:*

- *binary operation:*  $A \times A \rightarrow A$ ,  $(a, b) \mapsto ab$ .
- *bilinear:* *this operation is linear in each variable. In other words,  $\forall \alpha \in F$  and  $a, a', a'', b, b', b'' \in A$ , we have:*

$$(\alpha a)b = a(\alpha b) = \alpha(ab)$$

$$(a' + a'')b = a'b + a''b$$

$$a(b' + b'') = ab' + ab''$$

*The algebra  $A$  is associative if  $\forall a, b, c \in A$ ,  $(ab)c = a(bc)$ .*

**Example 1.0.2.**  $M_n(F)$ , the algebra of  $n \times n$  matrices over  $F$ .

**Example 1.0.3.** Let  $V$  be a vector space,  $\text{End}_F(V) = \{\text{linear transformations } \varphi : V \rightarrow V\}$  is an associative algebra with the composition operation  $(\phi\psi)(v) = \phi(\psi(v))$ . If  $\dim V < \infty$ , when we fix a basis, then we get a matrix representation of  $\varphi$  in our fix basis.

The most important thing everyone should know is Wedderburn's theorem.

**Definition 1.0.4.** Suppose  $F$  is algebraically closed. An algebra  $A$  is simple if

1.  $\nexists I \triangleleft A$  and  $I \neq 0$ .
2.  $A \cdot A \neq (0)$ .

**Corollary 1.0.5.** Any simple finite dimensional associative algebra is isomorphic to  $M_n(F)$  for some  $n$ .

**Corollary 1.0.6.** Radical

## 1.1 Where do Lie algebras come from?

**Definition 1.1.1.** A Lie algebra is a vector space  $L$  over a field  $F$ , with a binary operation  $[a, b] \in L$ , such that

1.  $[a, b]$  is bilinear.
2.  $[a, b] = -[b, a]$ .
3.  $[a, [b, c]] + [c, [a, b]] + [b, [c, a]] = 0$ , this is called Jacobi identity.

### 1.1.1 They come from associative algebras

Let  $A$  be an associative algebra.  $\forall a, b \in A$ , define  $[a, b] = ab - ba$ . Check that  $(A, [ , ])$  is a Lie algebra:

*Proof.* Bilinearity and skew-symmetry are clear. For Jacobi identity:

$$\begin{aligned}
 & [a, [b, c]] + [b, [c, a]] + [c, [a, b]] \\
 &= a(bc - cb) - (bc - cb)a + b(ca - ac) - (ca - ac)b + c(ab - ba) - (ab - ba)c \\
 &= abc - acb - bca + cba + bca - bac - cab + acb + cab - cba - abc + bac \\
 &= 0.
 \end{aligned}$$

□

We denote this new algebra  $(A, [ , ])$  by  $A^{(-)}$ .

Suppose that  $L \subseteq A$  and  $[L, L] \subseteq L$ . Then  $L \leq A^{(-)}$

**Example 1.1.2.**  $M_n(F)^{(-)} = \mathfrak{gl}(n)$ . We call it the general linear Lie algebra.

$[L, L] = \{\sum_i [a_i, b_i]\} \subseteq L$ . Clearly  $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq \mathfrak{gl}(n)$  and for  $n \geq 2$ ,  $[\mathfrak{gl}(n), \mathfrak{gl}(n)] \neq 0$ .

**Example 1.1.3.** Focus on  $[\mathfrak{gl}(n), \mathfrak{gl}(n)]$ , all matrices are of trace 0. This is called the special linear Lie algebra, denoted by  $\mathfrak{sl}(n)$ .

**Definition 1.1.4.**  $Z$  is the center of Lie algebra  $L$  if  $Z = \{z \in L \mid [z, L] = 0\}$ .

We can also find Lie algebras from the involution of an associative algebra. Let  $A$  be an associative algebra.  $*$  :  $A \rightarrow A$  is a linear transformation.  $*$  is called an involution if  $(a^*)^* = a$  and  $(ab)^* = b^*a^*$ ,  $\forall a, b \in A$ .

# Chapter 2

## Lecture 2

Recall last time:

1. Matrix Lie algebras, e.g.  $\mathfrak{sl}(n)$ . Consider involution  $*$  :  $M_n(F) \rightarrow M_n(F)$  skew-symmetric matrices.

2. Derivation,  $\text{Der}(A)$ . e.g.  $A = F[t_1, \dots, t_n]$ ,  $d = \sum_{i=1}^n f_i(t) \partial_{t_i}$

A non-associative algebra,  $d : x \mapsto [a, x]$  is a derivation.

Let  $L$  a Lie algebra, inner derivation

Q:  $d$  a derivation and  $\exp(d)$ . It need convergence.

**Example 2.0.1.**  $d = t_1 t_2 \frac{\partial}{\partial t_1}$ .

3. Brackets:

### 2.1 Engel Theorem

$L^n$ : span of products of elements of length  $n$ . Recall  $\text{ad}([a, b]) = [\text{ad}(a), \text{ad}(b)]$ .  $L \rightarrow \text{End}_F(L)^{(-)}, a \mapsto \text{ad}(a)$ .

$[I, J]$  is again an ideal.

Decending chain:

$$L = L^1 \supseteq L^2 \supseteq \dots$$

$L$  is nilpotent if...

eg: strict upper matrix

Another decending chain:  $L^{[1]} = [L, L]$ ,  $L^{[n+1]} = [L^{[n]}, L^{[n]}]$

$L$  is solvable if .....

e.g. 2 dimensional solvable non nilpotent Lie alg.

$A$  associa.  $L \subseteq A^{(-)}$ . Suppose  $A = \langle L \rangle$  as an associa alge. Then  $A$  is an enveloping algebra of  $L$ .

**Theorem 2.1.1.** Let  $A$  be a finite dimensional associative algebra,  $L \subseteq A^{(-)}$ ,  $A = \langle L \rangle$ .  $\exists n \geq 1, \forall a \in L, a^n = 0$ . Then  $A$  is nilpotent.

We say that a subset  $S \subseteq L$  is a Lie subset if  $\forall a, b \in S, [a, b] \in S$ .

**Theorem 2.1.2.** *Let  $A$  be a finite dimensional associative algebra,  $L \subseteq A^{(-)}$ ,  $A = \langle L \rangle$ . Let  $S$  be a Lie subset of  $L$ .  $L = \text{span}(S)$ ,  $\forall a \in S, a^n = 0$ .*

Choose a maximal Lie subset  $S_1 \subset S$  such that the associative  $\langle S_1 \rangle$  is nilpotent. So  $\exists n$  such that  $S_1 \dots S_1 = (0)$  ( $n$  times) (existence?)  
 claim  $\text{ad}(S_1) \dots \text{ad}(S_n) = (0)$ .